✱ Probability Distribution.

① Descrete Prob. Dist.

② Continous Prob. Dist.

① Descrete prob. Dist. –

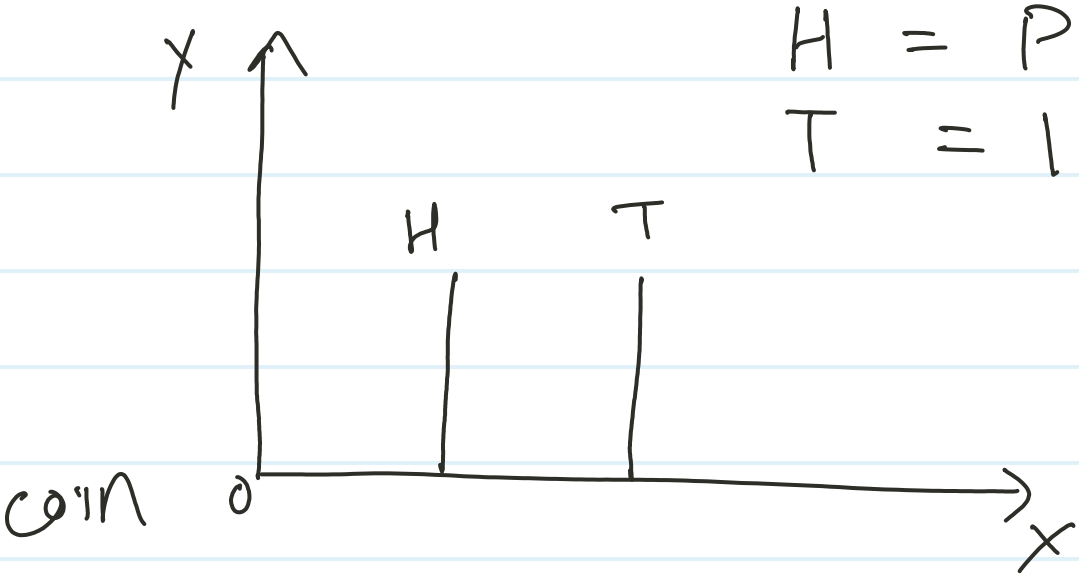
Ⅰ Bernaulli's Dist.

experiment = single

No. of output = fix

$H = P$

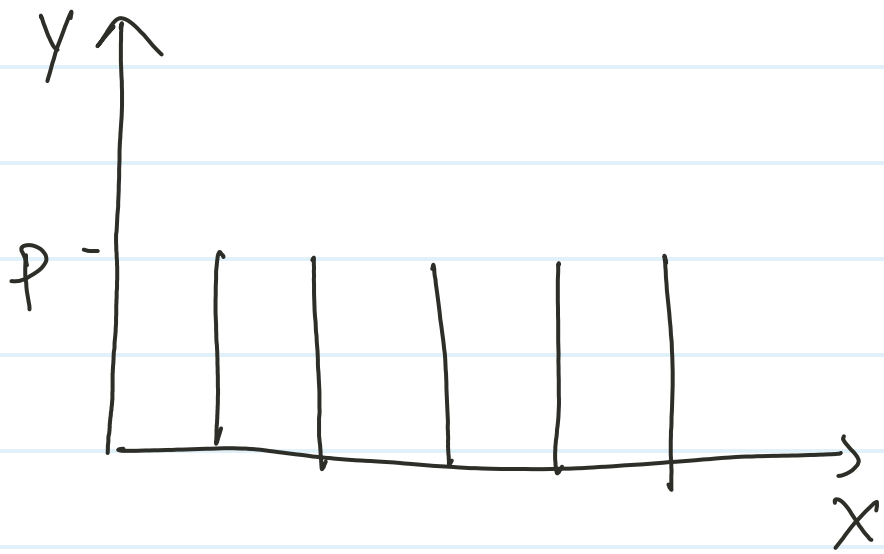
$T = 1 - P$



(II) Binomial Dist. -

experiment = more than one

no of outcome = fix



$H = p$

$T = 1-p$

$$P(X=r) = {}^{n}C_{r}\, p^{r} (1-p)^{n-r}$$

(III) poisson Distribution –

$$P(X = x) = \frac{e^{-\lambda} \lambda^{x}}{x!}$$

λ = Avg. no of time the event has occured in a certain period.

X = no. of events in that time interval.

e = Euler's num. $(e = 2.71818)$

$$\Rightarrow \frac{e^{-10} \cdot 10^{5}}{5 \times 4 \times 3 \times 2 \times 1}$$

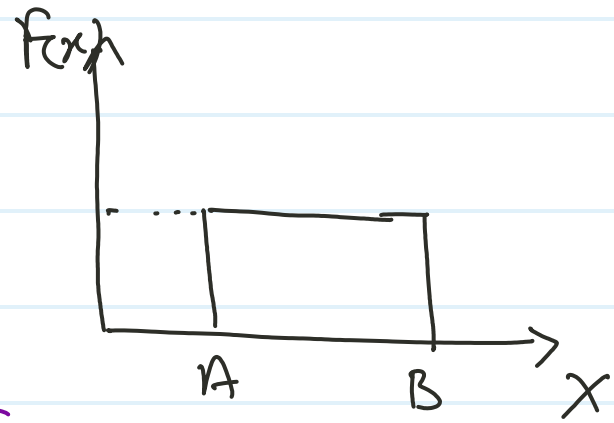
$\Rightarrow 0.037$

(II) Continous Probability Dist.

(I) uniform Distribution –

Eg:- Dice $= \frac{1}{6}$

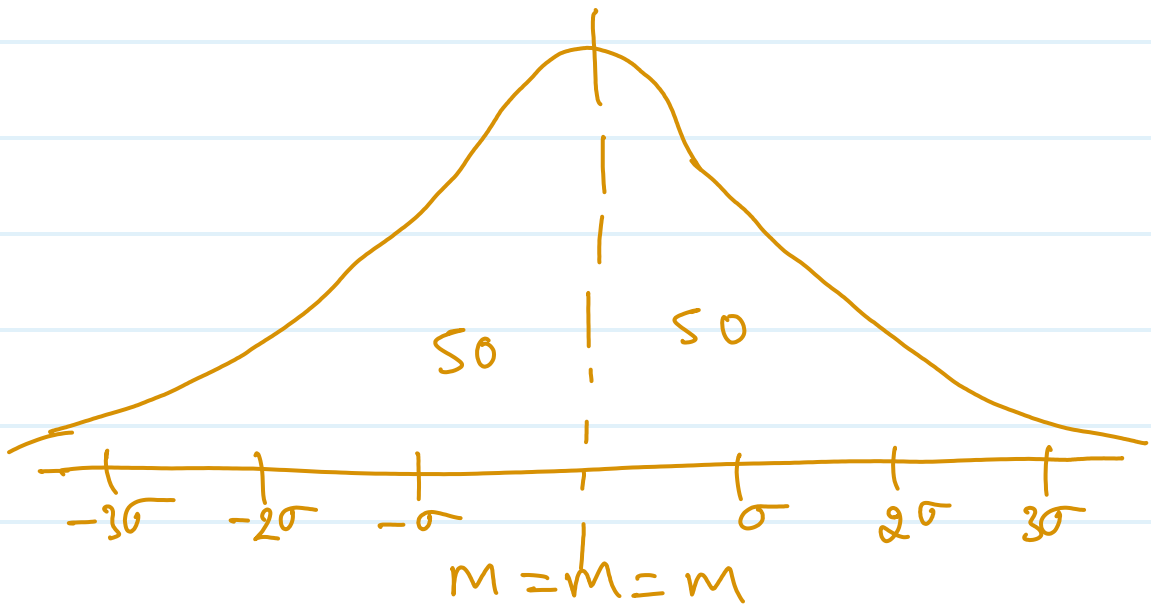
Card $= \frac{1}{52}, \frac{13}{52}, \frac{4}{52}$



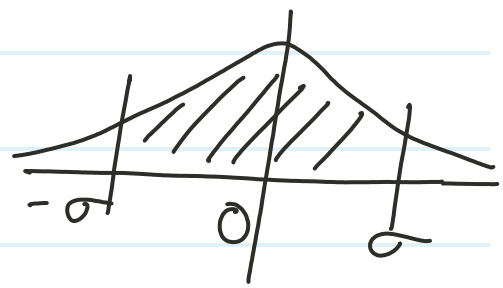
mean $E(x) = \frac{a+b}{2}$

variance $V(x) = \frac{(b-a)^2}{2}$

(II) Normal Dist. | Gaussian Dist. | Bell curv.



(III) standard normal Dist.



SD = 1 mean = 0

$$\text{Zscore} = \frac{x_i - \mu}{\sigma}$$

$\Rightarrow$ [1, 2, 3, 4, 5, 6, 7, 8, 9]

$\mu = \frac{45}{9} = 5$, $\sigma = 1$ (Assumed)

$x_i - \mu$

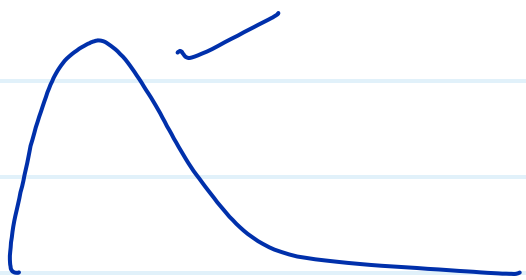
1 - 5	=	-4	-4/1	= -4
2 - 5	=	-3	-3/1	= -3
3 - 5	=	-2	-2/1	= -2
4 - 5	=	-1	-1/1	= -1
5 - 5	=	0	0	= 0

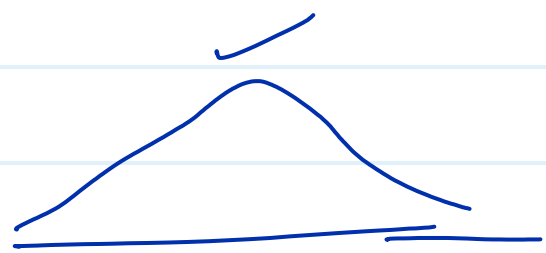
Age	Salary	exp.
25	25K	1
30	50k	3
40	1L.	6
60	2L.	25
80	10k.	30

18 – 80

10k – 2L.

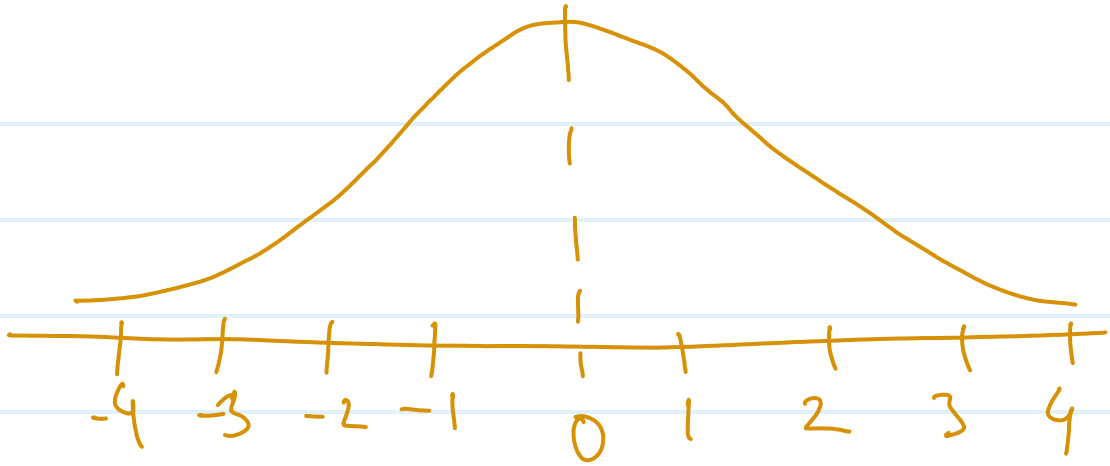
1 – 30





log(1, 2, 3, 4, 5, 6, 7, 8, 9)

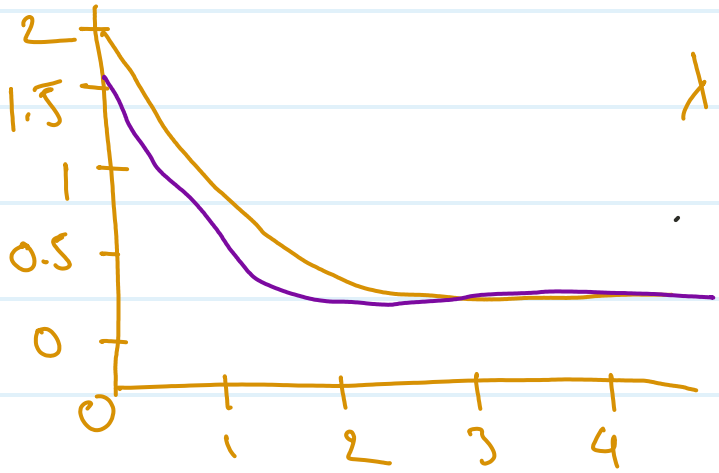
Guessing



(IV) Log normal Dist. / Right skewed / Positive Dist

mean > median > mode

(V) Exponential Dist. –



$\lambda = 1$

2 calls per hour.

1 calls 30 minute / ½ hour.

$\lambda = 0.5$

$f(x) = \lambda e^{-\lambda x}$

mean $E(x) = \frac{1}{\lambda}$

variance $V(x) = \left(\frac{1}{\lambda}\right)^2$

Power law Dist. / Pareto dist.

20 – 80 rule.

Cricket

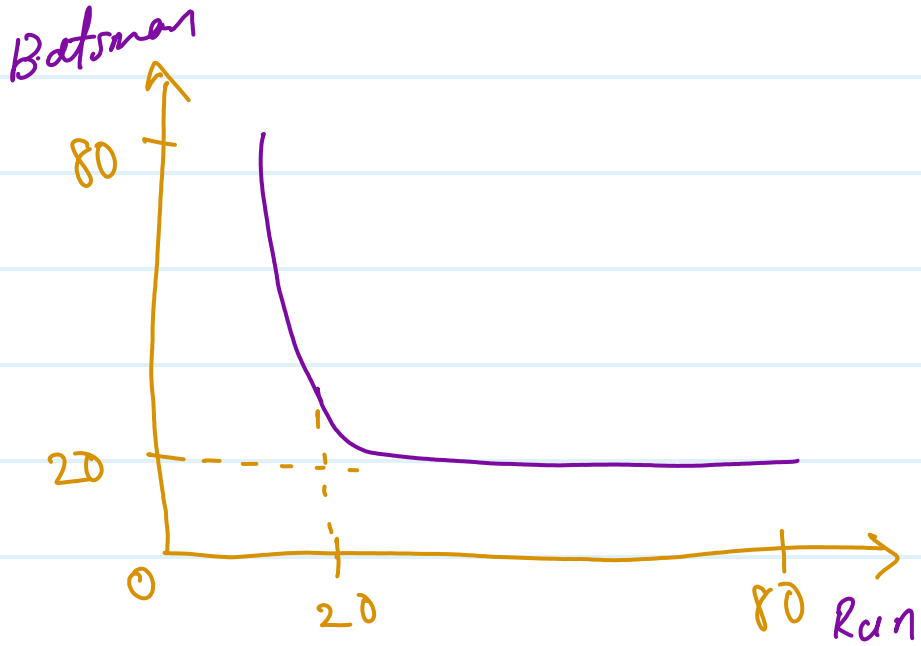
80% run – 20% batsman

20% run – 80% batsman

Family income

80% – 20%

20% – 80%

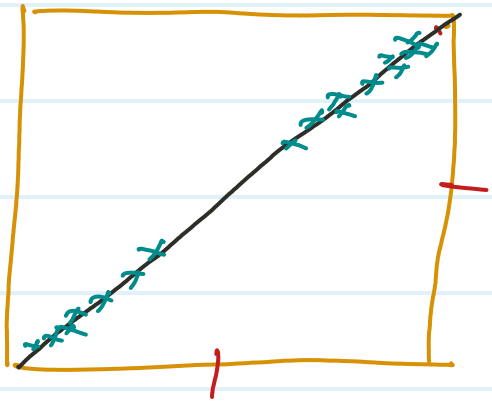


To transform in normal dist. we use two method

① Box-cox transformation
② Log normal transformation.

☆ Quartile-Quartile plot (Q-Q plot)

This is non-Gaussian Distribution.



It is normal / Gaussian Distribution.

X
60
70
65
90

$\mu = 71.25$

$\sigma = 10$

$$\frac{x_i - \mu}{\sigma}$$

$x_i - \mu$

$60 - 71.25 = -11.25$

$70 - 71.25 = -1.25$

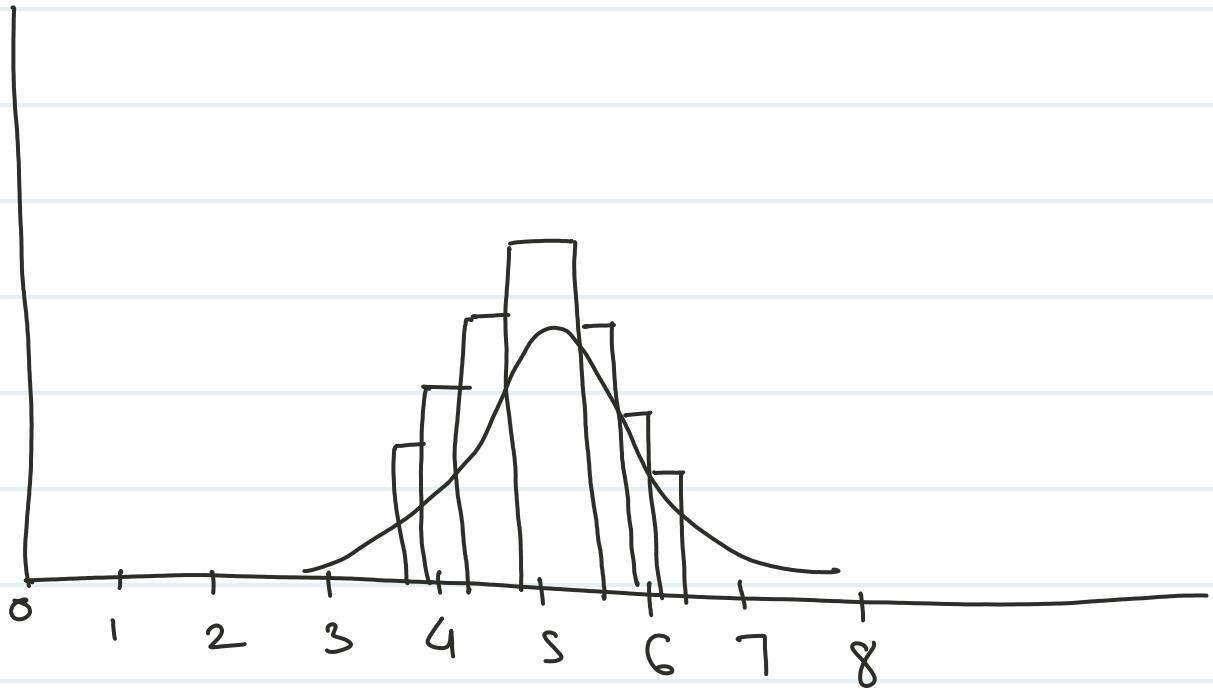
$65 - 71.25 = -6.25$

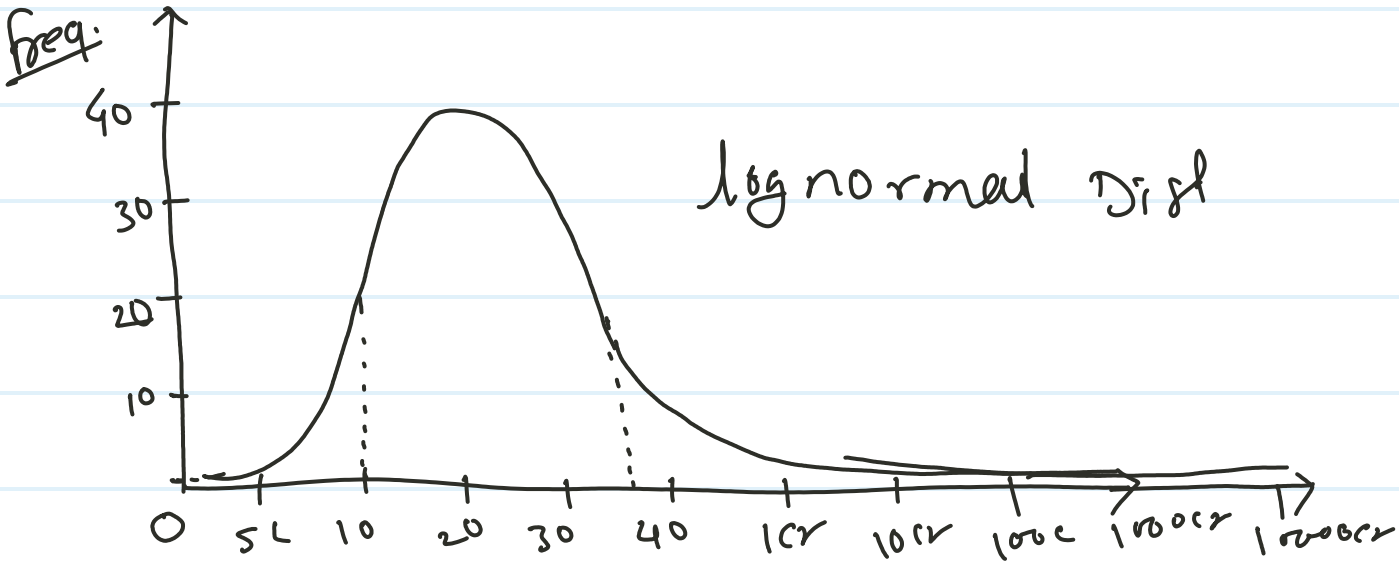
$90 - 71.25 = 18.75$

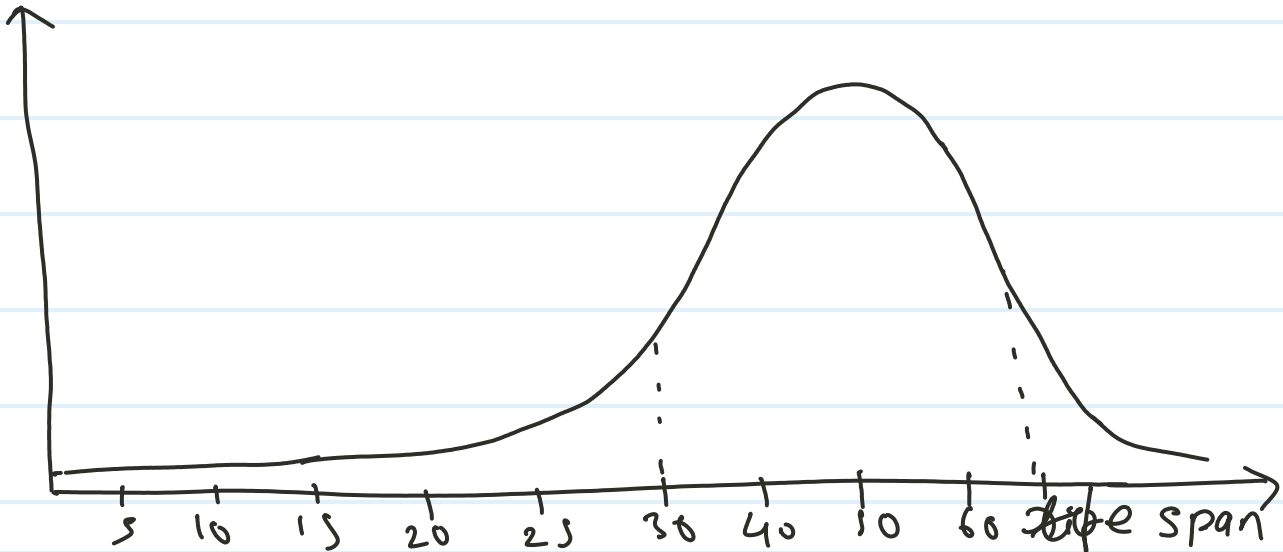
$$\frac{(-11.25) + (-1.25) + (-6.25) + 18.75}{10}$$

① -1.125 ③ -0.625

② -0.125 ④ 1.875


0
1
2
3
4
5
6
7
8


freq.
40
30
20
10
lognormal Dist
0
5L
10
20
30
40
1Cr
10Cr
100C
1000Cr
10000Cr


5
10
15
20
25
30
40
50
60
life span

Years	cm	Rup	Rup
Age	height	income	saving.
18	150	15000	10000000
30	170	30000	5000000
45	165	50000	20000000
60	180	100000	50000000
70	190	500000	1 cr.

$$\frac{18}{10000000} = 0.0000018$$

$$\frac{170}{10000000} = 0.000017$$

$$\frac{500000}{10000000} = 0.05$$

$$\frac{9900000}{10000000} \quad \underline{\underline{0.99}}$$

$$\frac{10000000}{10000000} \quad 1$$

X_1

10

20

30

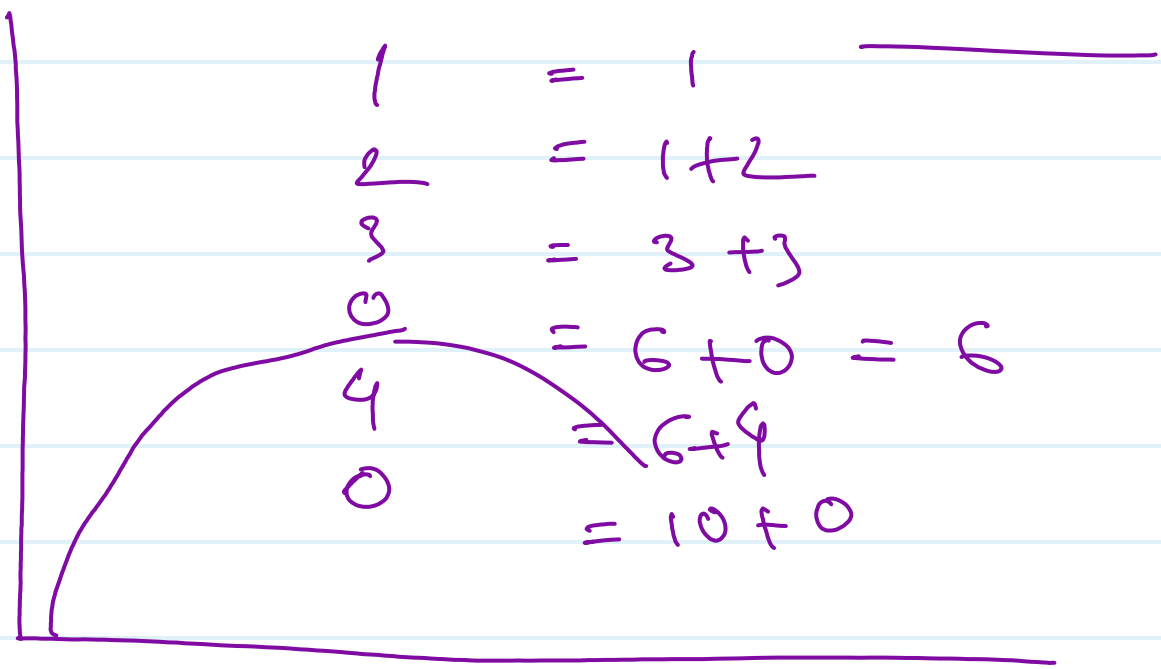
40

$$= \frac{X - X_{min}}{X_{max} - X_{min}}$$

$$= \frac{10 - 10}{40 - 10} = 0$$

$$= \frac{20 - 10}{40 - 10} = \frac{10}{30} \Rightarrow 0.3$$

1000

$$= [0, 1]$$

$$= \frac{1000 - 10}{1000 - 10} \Rightarrow \frac{\cancel{990}}{\cancel{990}} \; 1$$